

## Chapter 3, Section 7

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1. Let  $X$  be an integral projective scheme of dimension  $\geq 1$  over a field  $k$ , and let  $\mathcal{L}$  be an ample invertible sheaf on  $X$ . Then  $H^0(X, \mathcal{L}^{-1}) = 0$ .

*Proof.* A global section of  $\mathcal{L}^{-1}$  defines an  $\mathcal{O}_X$ -module morphism

$$\varphi : \mathcal{L} \rightarrow \mathcal{O}_X,$$

and hence induces morphisms

$$\varphi^{\otimes n} : \mathcal{L}^{\otimes n} \rightarrow \mathcal{O}_X$$

for all  $n > 0$ . Since  $\mathcal{L}$  is locally free of rank one,  $\varphi$  is locally given by multiplication by a section of  $\mathcal{O}_X$ . If for some  $x \in X$  the stalk map  $\varphi_x^{\otimes n}$  is not injective, it must be the zero map. As  $X$  is integral, this forces  $\varphi$  to vanish at the generic point, hence  $\varphi = 0$ . Assume therefore  $\varphi \neq 0$ . Since the global section functor is left exact,  $\varphi^{\otimes n}$  induces an injective map

$$\Gamma(\varphi^{\otimes n}) : H^0(X, \mathcal{L}^{\otimes n}) \rightarrow H^0(X, \mathcal{O}_X).$$

However,  $H^0(X, \mathcal{O}_X)$  is finite dimensional over  $k$  (5.2), while for  $n \gg 0$ , we have

$$\dim_k H^0(X, \mathcal{L}^{\otimes n}) = \chi(\mathcal{L}^{\otimes n}),$$

which is a polynomial in  $n$  (5.3), and hence eventually exceeds  $\dim_k H^0(X, \mathcal{O}_X)$ . This is a contradiction, and therefore  $\varphi = 0$ .  $\square$

2. Let  $f : X \rightarrow Y$  be a finite morphism of projective schemes of the same dimension over a field  $k$ , and let  $\omega_Y^\circ$  be a dualizing sheaf for  $Y$ .

- (a) Show that  $f^! \omega_Y^\circ$  is a dualizing sheaf for  $X$ , where  $f^!$  is defined as in (Ex. 6.10).
- (b) If  $X$  and  $Y$  are both nonsingular, and  $k$  algebraically closed, conclude that there is a natural trace map  $t : f_* \omega_X \rightarrow \omega_Y$ .

*Proof.*

- (a) We must first define a trace map  $t_X : H^n(X, f^! \omega_Y^\circ) \rightarrow k$ . By (Ex. 4.1), we have a natural isomorphism

$$H^n(X, f^! \omega_Y^\circ) \xrightarrow{\sim} H^n(Y, f_* f^! \omega_Y^\circ).$$

There is a natural morphism of  $\mathcal{O}_Y$ -modules  $f_* f^! \omega_Y \rightarrow \omega_Y$ , which induces a map between cohomology

$$H^n(Y, f_* f^! \omega_Y^\circ) \rightarrow H^n(Y, \omega_Y^\circ).$$

Composing with the trace map of  $\omega_Y^\circ$  gives us a trace map for  $f^! \omega_Y^\circ$ . It remains to show  $t_X$  satisfies the universal property, so let  $\mathcal{F}$  be a coherent sheaf on  $X$ . Then  $f_* \mathcal{F}$  is also a coherent sheaf on  $Y$ , so by (Ex. 6.10), we have

$$\mathrm{Hom}_X(\mathcal{F}, f^! \omega_Y) \cong \mathrm{Hom}_Y(f_* \mathcal{F}, \omega_Y) \cong H^n(Y, f_* \mathcal{F})^\vee \cong H^n(X, \mathcal{F})^\vee.$$

- (b) By (Ex. 6.10), defining a morphism  $t : f_* \omega_X \rightarrow \omega_Y$  is the same as defining a morphism  $\omega_X \rightarrow f^! \omega_Y$ , so we can let  $t$  be the morphism corresponding to a fixed (unique) isomorphism  $\omega_X \rightarrow f^! \omega_Y$ .  $\square$

3. Let  $X = \mathbb{P}_k^n$ . Show that  $H^q(X, \Omega_X^p) = 0$  for  $p \neq q$ ,  $k$  for  $p = q$ ,  $0 \leq p, q \leq n$ .

*Proof.* Consider the Euler sequence of (II, 8.13)

$$0 \longrightarrow \Omega_X \longrightarrow \mathcal{O}_X(-1)^{\oplus n+1} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

For each  $0 \leq p \leq n$ , let  $\mathcal{G}^{p+1} = \bigwedge^{p+1} \mathcal{O}_X(-1)^{\oplus n+1}$ , which is isomorphic to the direct sum of  $\binom{n+1}{p+1}$  copies of  $\mathcal{O}_X(-1)$ . By (II, Ex. 5.16c), for each  $0 \leq p \leq n$ , there exists a filtration  $\{\mathcal{F}^q\}$  of  $\mathcal{G}^{p+1}$  such that

$$\mathcal{F}^q / \mathcal{F}^{q+1} \cong \Omega_X^q \otimes \bigwedge^{p+1-q} \mathcal{O}_X = \begin{cases} \Omega_X^q & q = p, p+1 \\ 0 & \text{otherwise.} \end{cases}.$$

Since  $\mathcal{O}_X$  has rank one,

$$\mathcal{G}^{p+1} = \mathcal{F}^0 = \mathcal{F}^1 = \dots = \mathcal{F}^p, \quad \mathcal{F}^p / \mathcal{F}^{p+1} \cong \Omega_X^p, \quad \mathcal{F}^{p+1} \cong \Omega_X^{p+1},$$

which can be put into the short exact sequence

$$0 \longrightarrow \Omega_X^{p+1} \longrightarrow \mathcal{G}^{p+1} \longrightarrow \Omega_X^p \longrightarrow 0.$$

Taking the long exact sequence of cohomology gives

$$\dots \longrightarrow H^i(X, \mathcal{G}^{p+1}) \longrightarrow H^i(X, \Omega_X^p) \longrightarrow H^{i+1}(X, \Omega_X^{p+1}) \longrightarrow H^{i+1}(X, \mathcal{G}^{p+1}) \longrightarrow \dots$$

Direct sums commute with cohomology (2.9), and  $H^i(X, \mathcal{O}_X(-1)) = 0$  for all  $i$ , which means  $H^i(X, \mathcal{G}^{p+1}) = 0$ , so from the long exact sequence, we obtain the isomorphisms

$$H^i(X, \Omega_X^p) \cong H^{i+1}(X, \Omega_X^{p+1}).$$

Hence, if  $q \geq p$ ,

$$H^q(X, \Omega_X^p) \cong H^{q-p}(X, \mathcal{O}_X) = \begin{cases} k & p = q, \\ 0 & \text{otherwise.} \end{cases}$$

The other Hodge numbers are obtained via Serre duality. □

**4. The Cohomology Class of a Subvariety.** Let  $X$  be a nonsingular projective variety of dimension  $n$  over an algebraically closed field  $k$ . Let  $Y$  be a nonsingular subvariety of codimension  $p$  (hence dimension  $n - p$ ). From the natural map  $\Omega_X \otimes \mathcal{O}_Y \rightarrow \Omega_Y$  of (II, 8.12) we deduce a map  $\Omega_X^{n-p} \rightarrow \Omega_Y^{n-p}$ . This induces a map on cohomology  $H^{n-p}(X, \Omega_X^{n-p}) \rightarrow H^{n-p}(Y, \Omega_Y^{n-p})$ . Now  $\Omega_Y^{n-p} = \omega_Y$  is a dualizing sheaf for  $Y$ , so we have the trace map  $t_Y : H^{n-p}(Y, \Omega_Y^{n-p}) \rightarrow k$ . Composing, we obtain a linear map  $H^{n-p}(X, \Omega_X^{n-p}) \rightarrow k$ . By (7.13) this corresponds to an element  $\eta(Y) \in H^p(X, \Omega_X^p)$ , which we call the *cohomology class* of  $Y$ .

- (a) If  $P \in X$  is a closed point, show that  $t_X(\eta(P)) = 1$ , where  $\eta(P) \in H^n(X, \Omega_X^n)$  and  $t_X$  is the trace map.
- (b) If  $X = \mathbb{P}^n$ , identify  $H^p(X, \Omega_X^p)$  with  $k$  by (Ex. 7.3), and show that  $\eta(Y) = (\deg Y) \cdot 1$ , where  $\deg Y$  is its *degree* as a projective variety (I, §7).
- (c) For any scheme  $X$  of finite type over  $k$ , we define a homomorphism of sheaves of Abelian groups  $d \log : \mathcal{O}_X^* \rightarrow \Omega_X$  by  $d \log(f) = f^{-1} df$ . Here  $\mathcal{O}_X^*$  is a group under multiplication, and  $\Omega_X$  is a group under addition. This induces a map on cohomology  $\text{Pic } X = H^1(X, \mathcal{O}_X^*) \rightarrow H^1(X, \Omega_X)$  which we denote by  $c$ .
- (d) Returning to the hypotheses above, suppose  $p = 1$ . Show that  $\eta(Y) = c(\mathcal{L}(Y))$ , where  $\mathcal{L}(Y)$  is the invertible sheaf corresponding to the divisor  $Y$ .